

HiVT: Hierarchical Vector Transformer for Multi-Agent Motion Prediction

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Abstract

Accurately predicting the future motions of surrounding traffic agents is critical for the safety of autonomous vehicles. Recently, vectorized approaches have dominated the motion prediction community due to their capability of capturing complex interactions in traffic scenes. However, existing methods neglect the symmetries of the problem and suffer from the expensive computational cost, facing the challenge of making real-time multi-agent motion prediction without sacrificing the prediction performance. To tackle this challenge, we propose Hierarchical Vector Transformer (HiVT) for fast and accurate multi-agent motion prediction. By decomposing the problem into local context extraction and global interaction modeling, our method can effectively and efficiently model a large number of agents in the scene. Meanwhile, we propose a translation-invariant scene representation and rotation-invariant spatial learning modules, which extract features robust to the geometric transformations of the scene and enable the model to make accurate predictions for multiple agents in a single forward pass. Experiments show that HiVT achieves the state-of-the-art performance on the Argoverse motion forecasting benchmark with a small model size and can make fast multi-agent motion prediction.

1. Introduction

Navigating through dynamic environments in a safe manner is a critical mission of autonomous vehicles. To this end, autonomous vehicles need to understand the surroundings and anticipate the future on the road. However, it is challenging to accurately predict the future motions of nearby traffic agents, such as vehicles, bicycles, and pedestrians, whose goals or intents may be unknown. In multi-agent traffic scenarios, an agent’s behavior is shaped by complex interactions with other agents. Such interactions further intertwine with the map-dependent traffic regulations, making it extremely difficult to understand the diverse

behavior of multiple agents in the scene.

Recently, learning-based methods have demonstrated their effectiveness in the motion prediction task [9, 12, 31, 32, 36, 49]. Inspired by the progress in computer vision, some works rasterize the scenes into bird’s eye view images and apply CNNs to make predictions [9, 12, 25]. Although these approaches are easy to implement with off-the-shelf image models, they are computationally expensive and have limited receptive fields. Given these limitations, recent works [17, 31, 49] employ a vectorized approach for more compact scene representations, which extracts a set of vectors or points from the trajectories and the map elements. The scenes are then processed by graph neural networks [6, 20, 29], Transformers [46], or point cloud models [39, 40, 47] to learn the relationships among vectorized entities such as trajectory waypoints and lane segments.

Existing vectorized approaches, however, are challenged by the need to make real-time motion predictions in rapidly changing traffic conditions. Since vectorized methods are generally not robust to translation and rotation of the reference frame, to mitigate this problem, recent research normalizes the scene to be centered at the target agent and to be aligned with the target agent’s heading [17, 31, 49]. This remedy becomes problematic when a large number of agents in the scene need to be predicted, owing to the expensive cost of re-normalizing the scene and re-computing the scene features for each target agent. Further, existing works model *all-to-all* relationships in the space and the time dimensions to capture fine-grained interactions among the vectorized entities [38, 51], which inevitably leads to prohibitive computation with the increase of entities. As making accurate predictions in real time is critical for the safety of autonomous driving, we are thus motivated to push the state-of-the-art by developing a new framework to achieve faster and more accurate multi-agent motion prediction.

In a nutshell, our approach exploits the symmetries and the hierarchical structure in the problem of multi-agent motion prediction. We frame the motion prediction task in multiple stages and hierarchically model the relationships

between entities based on Transformers [46]. *In the first stage*, our framework avoids expensive all-to-all interaction modeling and extracts context features only *locally*. Specifically, we divide the scene into a set of local regions, where each local region is centered at one modeled agent. For each agent-centric local region, we extract context features from the local vectorized entities which contain rich information related to the central agent. *In the second stage*, to compensate for the restricted local receptive fields and capture long-range dependencies in the scene, we perform global message passing among agent-centric local regions by empowering the Transformer encoder with the geometric relationships between local reference frames. *Finally*, given the local and the global representations, a decoder produces future trajectories for all agents in a *single* forward pass. To further leverage the symmetries of the problem, we employ a scene representation that is agnostic about the translation of the global coordinate frame, in which we use relative positions to characterize all vectorized entities. Based on this scene representation, we introduce rotation-invariant cross-attention modules for spatial learning, which can learn local and global representations that are invariant to the rotation of the scene.

Our approach has the following clear advantages. *First*, by decomposing the problem into local context extraction and global interaction modeling, our approach can progressively aggregate information at different scales and model a large number of entities in the scene with high efficiency. *Second*, our method can learn representations robust to translation and rotation of the inputs via a translation-invariant scene representation and rotation-invariant spatial learning modules. *Third*, our model can make faster and more accurate predictions with much fewer parameters than the state-of-the-art approaches. We validate all the above advantages via extensive experiments on large-scale driving data. Our code will be publicly available.

2. Related Work

Traffic Scene Representation. Tackling the motion prediction problem requires learning rich representations from the elements of the traffic scene, including the high-definition map and the past trajectories of agents. Extensive works have used rasterized scenes as model inputs [4, 9, 12, 15, 19, 25, 43] and employed standard image models [11, 24, 26, 44] for learning. Specifically, these methods extract map elements (e.g., lane boundaries, crosswalks, traffic lights) from the high-definition map and use different colors or masks to render the scenes as bird’s eye view images. The past trajectories of agents are either rasterized as additional image channels [9, 12] or processed by temporal models like RNNs [2, 41, 42].

Rasterized methods are compatible with the mature techniques in computer vision but are also costly and inefficient

in learning. Recently, vectorized methods [17, 31, 49] have gained popularity for their efficient sparse encoding and the capability of capturing complex structural information. Unlike the rasterized approaches, these methods treat the scene as a set of entities associated with semantic and geometric attributes and learn the relationships between entities. VectorNet [17] models the interactions among lane and trajectory polylines with graph neural networks. It is also used as the backbone network by some following works [22, 32, 52]. LaneGCN [31] builds a lane graph from the lane segments and exploits multi-scale graph convolutional networks to learn representations for graph nodes. TPCN [49] extends point cloud models to learn from the spatial-temporal point set composed of trajectory waypoints and lane points. Our scene representation also falls into this category but differs in that all vectorized entities are characterized by relative positions, making our representation *translation-invariant*.

Motion Prediction. Since social interactions are ubiquitous in traffic scenarios and significantly impact the future motions of traffic agents, many motion prediction approaches have considered the dependencies between agents’ behaviors and reasoned agent-agent interactions using social pooling [2, 13, 23], graph neural networks [8, 27, 31, 37], or attention mechanisms [17, 30, 32, 36, 38, 43, 48, 50]. Inspired by the success of Transformer models [46] in a wide range of domains [5, 7, 14, 16], some recent works employ Transformers in the motion prediction task to model spatial relationships, temporal dependencies, and relationships between agents and map elements [21, 30, 32, 36, 38, 50, 51]. In comparison, our Transformer architecture differs from existing ones by learning local and global representations hierarchically. This hierarchical strategy helps the model learn multi-scale features and is more efficient than those performing all-to-all message passing along the space and the time axes. Moreover, we model multiple agents via an agent-centric representation that is invariant to translation and rotation of the scene. The hierarchical architecture and the symmetric designs enable our method to achieve the state-of-the-art prediction performance with fewer parameters and less computational cost than other approaches.

3. Approach

3.1. Overall Framework

An overview of our proposed framework is illustrated in Fig. 1. We first organize a traffic scene as a collection of vectorized entities. Based on this scene representation, our framework hierarchically aggregates spatial-temporal information in the scene. In the first stage, we encode rotation-invariant local context features for each agent. The aggregation of ego-motion, neighboring agents’ motions, and local map structure can provide rich information related to the

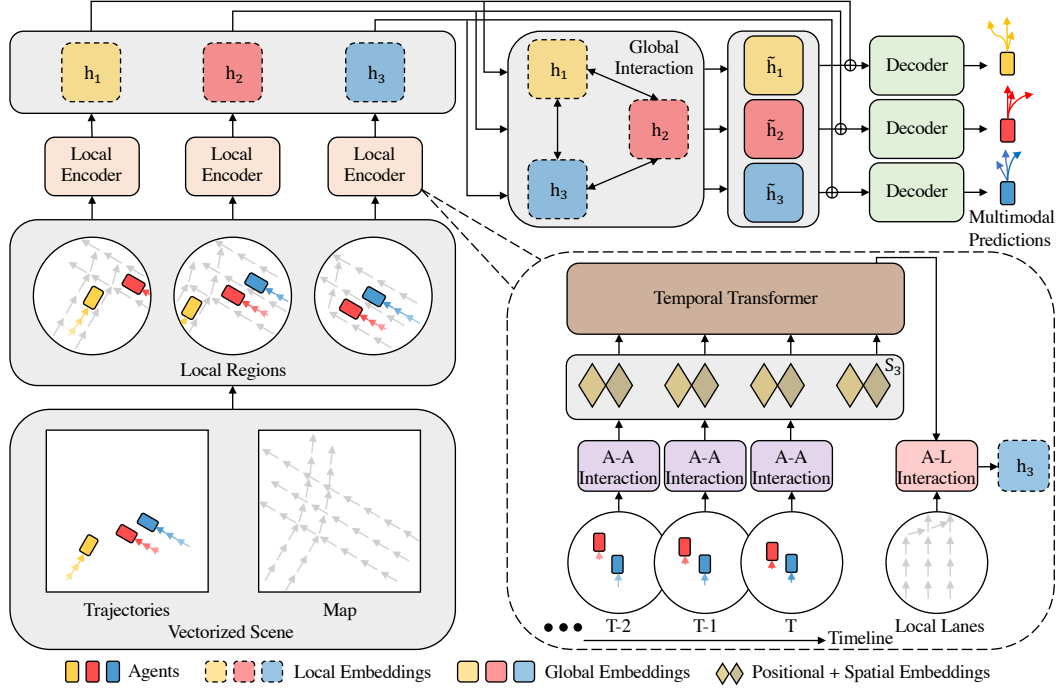


Figure 1. Overview of HiVT. A-A Interaction and A-L Interaction denote Agent-Agent and Agent-Lane Interaction respectively.

modeled agent. In the second stage, a global interaction module aggregates the local context of different agents and updates each agent’s representation to capture long-range dependencies and scene-level dynamics. Finally, the hierarchically learned representations are used to make multi-modal trajectory predictions for all agents simultaneously.

3.2. Scene Representation

A traffic scene consists of agents and map information. To represent the scene in a structured way, we extract vectorized entities from the scene, including the trajectory segments of traffic agents and the lane segments in map data. A vectorized entity is associated with semantic and geometric attributes. Compared with previous vectorized methods [17, 31, 49] in which the geometric attributes of agents or lanes involve the absolute positions of points, our representation avoids using any absolute positions and characterizes the geometric attributes with relative positions, which makes the scene a vector set entirely. Specifically, an agent i ’s trajectory is represented as $\{\mathbf{p}_i^t - \mathbf{p}_i^{t-1}\}_{t=1}^T$, where $\mathbf{p}_i^t \in \mathbb{R}^2$ is agent i ’s location at time step t and T is the total historical time steps. For a lane segment ξ , the geometric attribute is given by $\mathbf{p}_\xi^1 - \mathbf{p}_\xi^0$, where $\mathbf{p}_\xi^0 \in \mathbb{R}^2$ and $\mathbf{p}_\xi^1 \in \mathbb{R}^2$ are the starting and the ending coordinates of ξ . By transforming the point set into a vector set, such a representation guarantees translation invariance naturally. However, the information about the relative positions between entities is also discarded. To preserve spatial relationships, we intro-

duce relative position vectors for agent-agent and agent-lane pairs. For instance, the position vector of agent j relative to agent i at time step t is $\mathbf{p}_j^t - \mathbf{p}_i^t$, which fully depicts the spatial relationship of the two agents and is also translation-invariant. Without any loss of information, our scene representation ensures that any learnable functions applied to it will necessarily respect translation invariance.

3.3. Hierarchical Vector Transformer

To accurately predict the future trajectories of traffic agents in a highly dynamic environment, the model needs to effectively learn the spatial-temporal relationships among a large number of vectorized entities. Transformers [46] have shown promise in capturing long-range dependencies between entities in a variety of tasks [5, 7, 14, 16, 33]. However, directly applying Transformers to the spatial-temporal entities suffers from the complexity of $O((NT + L)^2)$, where N , T , and L are the numbers of agents, historical time steps, and lane segments, respectively. To efficiently learn from a large number of entities, our model factorizes the space and the time dimensions and learns spatial relationships only *locally* at each time step. Specifically, we divide the space into N local regions, and each local region is centered at one agent in the scene. Within each local region are the trajectory segments and the local environment of the central agent, where the environmental information involves the trajectory segments of the neighboring agents and the local lane segments surrounding the